

# A Multimodal Early Warning Pipeline for Cardiac Arrest Using Foundation Model Embeddings

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## Abstract

We present a complete, reproducible pipeline that scores the risk of cardiac arrest within the next 1, 6 or 24 hours from bedside monitor signals: electrocardiogram, photoplethysmogram, arterial blood pressure and heart rate. Two open foundation models, ECG FM (a wav2vec 2.0 style model with 90.9 million parameters) and PaPaGei S, summarize the raw waveforms into embeddings, and a temporal network in the Wav2Arrest style (bidirectional LSTM with learned attention) consumes a 6 hour history of fused features to produce per horizon risk estimates and a time to event estimate. The pipeline was validated end to end on a synthetic cohort of 60 patients with 28 arrest events, where it reached a 6 hour AUROC of 0.891 on held out patients with a first true alert 21.7 hours before arrest. A stress test on the open access Sudden Cardiac Death Holter Database produced a genuine negative result: the synthetic trained model does not transfer to real data, and no model variant trained on 20 real patients beats chance at 6 hours, including identity adversarial training and three alternative fusion schemes with explicit missing modality masks. At 1 hour a weak signal exists and comes from engineered rate features rather than the foundation model embeddings. We discuss why this outcome is expected for Holter data without hemodynamic signals, and identify the credentialed MIMIC III evaluation as the decisive next step. This report is a research prototype and makes no clinical claim.

## 1 Introduction

Cardiac arrest in hospital often follows hours of measurable hemodynamic deterioration, which makes early warning a promising target for machine learning on continuous monitor data. Wav2Arrest and its successor introduced a strong recipe for this task: a foundation model feature extractor, a temporal aggregator, time to event targets and patient level evaluation. We follow that recipe and extend it to four signal modalities.

The contribution of this work is an open, reproducible implementation of the full chain: data, labels, preprocessing, engineered features, embeddings from two real foundation models, a temporal risk model, an alert simulator and a strict evaluation protocol. We report results on a physiologically coupled synthetic cohort and on the only open access real pre arrest dataset, the Sudden Cardiac Death Holter Database. Both results are reported honestly: the synthetic numbers validate the machinery, and the real data test currently fails, which we analyze rather than hide.

## 2 Methodology

### 2.1 Pipeline overview

Figure 1 shows the pipeline. Signals are resampled to a common 125 Hz clock and cut into 5 minute windows. Each window receives three feature blocks: engineered features computed from the waveforms, an ECG FM embedding and a PaPaGei S embedding. The fused vector

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**Algorithm 1** Synthetic cohort generation

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```
1: Inputs: patient count  $N$ , arrest fraction  $p$ , stay length range, drift window  $D$ 
2: for each patient  $i$  in  $1 \dots N$  do
3:   draw stay length and admission time
4:   mark case with probability  $p$ ; if case, draw arrest time inside stay
5:   for each 5 min window  $w$  in stay do
6:     draw baseline vitals (heart rate, pressures, saturation) with wander and noise
7:     if case and window inside last  $D$  hours before arrest then
8:       apply drift: heart rate up, pressures and saturation down, transit time up
9:     end if
10:    if case and window after arrest then
11:      mark window excluded (post arrest)
12:    end if
13:    draw quality flag per stream (ok, flat, sat, noise)
14:  end for
15: end for
16: return vitals table, event table
```

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feeds a temporal network that watches the last 6 hours and outputs three risk scores plus a time to event estimate. Alerts fire when the 6 hour risk crosses a threshold, subject to a refractory period.

## 2.2 Synthetic cohort generation

The synthetic cohort produces monitor style signals with physiological coupling so that every later stage has a learnable, quality checked input. Beat trains are generated for the electrocardiogram (P, QRS and T waves as Gaussian bumps), and each R peak is followed by a photoplethysmogram pulse and an arterial pressure pulse delayed by the pulse transit time, which lengthens as blood pressure falls. Arrest patients receive a 6 hour drift: rising heart rate, falling blood pressure and oxygen saturation, falling heart rate variability and rising pulse transit time. Quality defects (flatline, saturation, noise bursts) are injected in five percent of stream windows, and ten percent of patients lack the photoplethysmogram stream. Algorithm 1 summarizes the generator.

## 2.3 Cohort and labels

For each horizon  $H \in \{1, 6, 24\}$  hours, a window receives a binary label that is 1 when the arrest occurs within the next  $H$  hours, together with a time to event target in hours for case patients. Windows during or after resuscitation are excluded, as are windows in the final 15 minutes before arrest, when clinicians are already responding. Excluded windows keep their row with a reason field so that downstream code can assert the exclusion. Algorithm 2 gives the procedure.

## 2.4 Preprocessing and quality control

All streams share one clock at 125 Hz. Resampling to encoder targets uses linear interpolation for upsampling (the ECG FM preprocessing does the same) and polyphase filtering for decimation. Each stream passes a rule based gate: flatline (standard deviation below threshold), saturation (fraction of samples pinned at the rails), and a signal to noise heuristic (physiological band power versus high frequency band power). Segments of 30 seconds are cut deterministically per patient and window, so embeddings can be cached and recomputed without drift. Algorithm 3

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**Algorithm 2** Window labeling and exclusions

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```
1: for each patient and each 5 min window do
2:   if window ends after arrest time then
3:     mark excluded, label missing ▷ post arrest
4:   else if window ends within 15 min of arrest then
5:     mark excluded, label missing ▷ response phase
6:   else
7:     for each horizon  $H$  in 1, 6, 24 hours do
8:       label is 1 if time to arrest is at most  $H$  hours, else 0
9:     end for
10:    time to event equals arrest time minus window end time (cases only)
11:  end if
12: end for
```

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**Algorithm 3** Per stream quality gate

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```
1: Inputs: segment  $x$ , rate  $fs$ 
2: if all samples missing then
3:   return fail (missing stream)
4: end if
5: if standard deviation of  $x$  below flat threshold then
6:   return fail (flatline)
7: end if
8: saturation flag if pinned fraction above threshold
9: signal power in 0.5 to 12 Hz band, noise power in 30 to 60 Hz band
10: low ratio flag if signal to noise ratio below threshold
11: return pass if no flag raised, else fail with flags
```

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states the gate. Splits are patient level at all times: no patient appears in more than one of train, validation and test.

## 2.5 Engineered features

Each window receives ten engineered features computed from its waveforms: heart rate variability (mean RR interval, SDNN, RMSSD), ectopy burden as the fraction of premature RR intervals, arterial pressure mean and variability, the pulse transit time estimated per beat as the delay from each R peak to the steepest upstroke of the following photoplethysmogram pulse (median over beats), a quality pass indicator, and 30 minute trends of heart rate and systolic pressure from the vitals table. On the synthetic cohort the estimated transit time correlates with the ground truth at 0.979. Algorithm 4 lists the computation.

## 2.6 Foundation model embeddings

Two frozen foundation models summarize the waveforms. ECG FM expects 12 leads at 500 Hz in 5 second segments and returns 768 dimensional embeddings; the single monitor lead is placed in the lead II slot and the remaining leads are zero filled, which mirrors reduced lead use at the bedside. PaPaGei S expects 10 second photoplethysmogram segments at 125 Hz and returns 512 dimensional embeddings. The two encoders require conflicting Python environments, so each runs in a dedicated conda environment invoked by subprocess, and every embedding is cached on disk. Segment embeddings are mean pooled to the 5 minute window level. Algorithm 5 describes the orchestration.

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**Algorithm 4** Engineered features per window

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```
1: detect R peaks in the 30 s electrocardiogram
2: RR intervals; mean RR, SDNN, RMSSD, ectopy fraction
3: arterial pressure mean and variability from the pressure waveform
4: for each R peak do
5:   take the photoplethysmogram window from 30 ms to 550 ms after the peak
6:   foot equals position of steepest upstroke in that window
7:   delay equals foot position in milliseconds
8: end for
9: transit time equals median delay over beats
10: join 30 minute heart rate and pressure trends from the vitals table
11: return ten feature values
```

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**Algorithm 5** Embedding extraction

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```
1: for each 5 min window do
2:   cut the 30 s segment, resample to encoder rate, z score
3:   electrocardiogram: map lead to slot II, zero fill, 5 s chunks
4:   photoplethysmogram: 10 s chunks
5: end for
6: for each encoder in (ECG FM, PaPaGei S) do
7:   run extractor via subprocess in its dedicated environment
8:   cache segment embeddings on disk
9: end for
10: mean pool segment embeddings per window
```

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## 2.7 Temporal risk model

The model follows the FEAN design of Wav2Arrest. Each window vector (7 vitals, 10 engineered, 768 electrocardiogram, 512 photoplethysmogram, 1297 in total) is projected to a hidden size, processed by a two layer bidirectional LSTM over the last 72 windows (6 hours) with packing for variable lengths, pooled by learned attention with a padding mask, and mapped by linear heads to three risk logits and a normalized time to event logit. Training minimizes weighted binary cross entropy per horizon (weights inverse to class frequency) plus mean squared error on the time to event head for positive windows. Algorithm 6 gives the forward pass and loss.

Two extensions address generalization and fusion. First, an identity adversarial head (gradient reversal against a patient ID classifier on the context vector, ramped in over five epochs) discourages the representation from encoding patient identity, optionally combined with PC-Grad, which projects conflicting task gradients onto each other’s normal planes. Second, a fusion comparison replaces the single concatenation projection with per modality projectors followed by one of: hierarchical gated fusion, per window modality attention, or static learned per modality weights. Each modality carries an explicit presence mask; absent blocks are routed to learned per modality missing embeddings instead of being treated as zero valued data. On the synthetic cohort the fusion variants match the concatenation baseline within noise (6 hour AUROC 0.77 to 0.86), and with the photoplethysmogram removed at evaluation time gated and weighted fusion remain stable while attention fusion degrades; the identity adversarial variants are sanity checked there as well (no NaNs, losses decrease, and the identity probe stays high by construction because case and control are patient level properties on synthetic data).

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**Algorithm 6** FEAN forward pass and training loss

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```
1: Forward:
2: project each window vector to hidden size, apply dropout
3: pack sequences, run two layer bidirectional LSTM
4: attention scores with padding masked to minus infinity
5: context vector equals attention weighted mean
6: for each horizon  $H$  in 1, 6, 24 do
7:   risk logit equals linear head applied to context
8: end for
9: time to event logit equals linear head applied to context
10: Loss:
11: weighted binary cross entropy with logits per horizon, summed
12: plus mean squared error on time to event for positive windows
```

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**Algorithm 7** Alert replay simulation

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```
1: for each window of a patient in time order do
2:   append standardized features to history, keep last 72 windows
3:   compute risk for the alert horizon
4:   if risk above threshold and time since last alert above refractory then
5:     raise alert, record time and risk
6:   end if
7: end for
```

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## 2.8 Alert simulation and evaluation

Evaluation streams each held out patient in time order, as a bedside monitor would. An alert fires when the 6 hour risk reaches 0.5, and no further alert may fire within 60 minutes (refractory period). Metrics are AUROC, AUPRC and Brier per horizon with patient level bootstrap confidence intervals (1000 resamples), PPV at fixed sensitivity, sensitivity at a fixed alert budget, alerts per patient day, lead time of true alerts, calibration and the time averaged AUC across lead time bins (0 to 1, 1 to 6, 6 to 24 hours). Algorithms 7 and 8 state the simulation and the bootstrap.

## 3 Results

### 3.1 Synthetic validation

Table 1 compares the baselines and the temporal model on the patient level validation split. Table 2 reports the MEWS style score built from the streams we have (heart rate, systolic pressure, saturation; respiratory rate, temperature and consciousness are missing and stated as such). Table 3 reports the held out replay evaluation, Table 4 the alert level metrics, and Table 5 the feature block ablations.

The synthetic results validate the machinery. They are inflated by design: the drift is deterministic, and the 1 hour horizon is trivially separable. The numbers must not be read as clinical evidence.

Figures 2 to 5 illustrate the synthetic regime. Figure 2 shows the pre arrest drift, Figure 3 one clean window and the same window with an injected noise artifact rejected by the quality gate, Figure 4 the held out risk trajectories with alerts, and Figure 5 the calibration curves.

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**Algorithm 8** Patient level bootstrap

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```
1: repeat
2:   resample patients with replacement
3:   restrict windows to sampled patients
4:   if both classes present then
5:     compute the metric on the restricted set
6:   end if
7: until 1000 iterations done
8: confidence interval equals the 2.5 and 97.5 percentiles
```

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Table 1: Model performance on the patient level validation split (AUROC).

| Model               | 1 h   | 6 h   | 24 h  |
|---------------------|-------|-------|-------|
| Logistic regression | 1.000 | 0.954 | 0.689 |
| Gradient boosting   | 0.998 | 0.993 | 0.911 |
| FEAN temporal model | 1.000 | 0.844 | 0.667 |

### 3.2 Real data stress test

The open access Sudden Cardiac Death Holter Database provides 23 two lead 250 Hz Holter recordings of 4 to 25 hours from patients who arrested during recording; 20 records carry a documented ventricular fibrillation onset time used as the arrest timestamp. We embedded 1829 five minute windows (the 24 hour horizon is not evaluable because no recording extends far enough before onset to supply 24 hour negatives) and ran a zero shot application of the synthetic trained model plus a 5 fold cross validation grouped by patient, for the baseline model, two identity adversarial variants, and three fusion variants with explicit missing modality masks. A protocol fix matters: in an earlier version, sequence histories crossed record and span boundaries, which confounded the results; all numbers below use the corrected protocol (histories stay inside one record and one span, fixed seeds) with patient level bootstrap confidence intervals. Table 6 reports the result.

We verified that this is a genuine negative result and not a training failure: train loss decreases across epochs while held out loss stays flat, and predictions spread widely, yet held out records remain at or below chance at 6 hours for every variant; on one fold several variants become confidently inverted (AUROC near zero), learning record specific patterns that do not transfer across 16 training patients. Figures 6 and 7 show one real patient trajectory and the learning curves.

Three variant level lessons. First, the models do recognize patients: a linear identity probe on the context vector classifies held out train windows into patients at 0.13 to 0.47 accuracy across variants (chance 0.05), so leakage is present and quantified. The identity signal is carried by the embedding block (probe 0.439 without engineered features versus 0.130 without embeddings). Identity adversarial training helps slightly at both horizons but does not reduce the probe (0.423 to 0.470), so gradient reversal does not remove the leakage; modality attention is the only architecture that substantially reduces it (probe 0.143), and it is also the best held out variant at 6 hours. Second, concatenation, gating and static weights are statistically indistinguishable; modality attention is the best deep variant at 6 hours but also the least stable. Missing modality masks keep predictions finite when streams disappear but do not change the ranking. Third, the 1 hour signal is carried by engineered heart rate variability and ectopy features (0.767 without embeddings), and the ECG FM embeddings dilute it (0.632 full model). Attention salience shows the model spreads its attention over history and puts near zero weight on the current window.

Table 2: MEWS style score alert statistics on the validation split.

| Threshold | Rate per day | 1 h   |      | 6 h   |      |
|-----------|--------------|-------|------|-------|------|
|           |              | Sens. | PPV  | Sens. | PPV  |
| 4         | 11.2         | 0.82  | 0.82 | 1.00  | 1.00 |
| 5         | 8.1          | 1.00  | 1.00 | 1.00  | 1.00 |

Table 3: Held out replay evaluation (two patients, patient level bootstrap CIs).

| Horizon | Positives | AUROC [95% CI]       | AUPRC | Brier |
|---------|-----------|----------------------|-------|-------|
| 1 h     | 12        | 1.000 [1.000, 1.000] | 1.000 | 0.025 |
| 6 h     | 72        | 0.891 [0.825, 0.922] | 0.785 | 0.315 |
| 24 h    | 288       | 0.946 [0.944, 0.948] | 0.976 | 0.137 |

### 3.3 Robustness

Five perturbation families were applied to the held out replay patients and scored with the trained models (concatenation versus mask aware gated and weighted fusion). Waveform perturbations were re embedded through the real encoders: ECG noise at SNR 20, 10 and 5 dB, baseline wander, 50 Hz mains, flatline halves, saturation, and PPG noise at 5 dB. Feature level stress covered vitals noise at 0.5 to 2 times the per column standard deviation, missing PPG, delayed measurements shifting the whole vector by 5, 10 and 15 minutes, and a sensor failure where PPG dies mid stay. Every perturbation lands within 0.03 of the clean 6 hour AUROC (0.929 to 0.945); the only directional effect is the delay family, which degrades monotonically but mildly (0.929 to 0.920 at 15 minutes). Figures 9 and 10 show the sweep.

Honest caveats: this measures robustness on synthetic data, where the risk signal lives in the vitals and engineered blocks (the earlier ablation finding), so waveform robustness partly reflects the encoders and partly that the embeddings matter little there; the quality gates would reject flatline and saturation windows in the real pipeline, and these tests bypassed the gates and still held; and with 2 replay patients these are point estimates.

## 4 Discussion

Three facts explain the negative real data result. First, only 16 training patients were available after grouping, far below what a temporal network needs to generalize; the identity adversarial head and three alternative fusion schemes did not change this, and a linear identity probe shows the representation still encodes patient identity, so the overfitting mechanism is not removed by gradient reversal. Second, the strongest modality blocks (photoplethysmogram, arterial pressure, vitals) were absent and zero filled; missing modality masks routed the absent photoplethysmogram to a learned embedding, which kept predictions finite but did not recover signal. Third, Holter recordings from the 1980s differ from ICU bedside monitors, and ventricular fibrillation onset often has no gradual electrocardiographic prodrome hours ahead, which is consistent with the literature. The one robust hint of signal, at 1 hour, comes from engineered rate features (heart rate variability, ectopy), not from embeddings, which suggests that what little premonitory information exists at that horizon is hemodynamic or rate based. This dataset therefore cannot validate a hemodynamic early warning system; it bounds the electrocardiogram only regime, and our results suggest that regime alone is not enough.

The decisive evaluation is the credentialed MIMIC III route: the Ext CA annotation set (36 arrest episodes from 31 ICU patients with photoplethysmogram plus electrocardiogram or arterial pressure) together with the waveform matched subset provides exactly the modalities

Table 4: Alert level metrics on the held out replay patients.

| Metric                                          | Value                           |
|-------------------------------------------------|---------------------------------|
| First true alert before arrest                  | 21.7 h                          |
| Lead time of true alerts (min, median, max)     | 0.6, 11.2, 21.7 h               |
| False alarms on the control patient             | 5 in 24 h (4.9 per patient day) |
| PPV at sensitivity 0.8 (1 h, 6 h, 24 h)         | 1.000, 0.307, 1.000             |
| Sensitivity at 1 alert per day (1 h, 6 h, 24 h) | 1.000, 1.000, 1.000             |
| Time averaged AUC across lead time bins         | 0.890                           |

Table 5: Feature block ablations (validation AUROC, 20 epochs each).

| Variant                | 1 h   | 6 h   | 24 h  | Replay alerts (case, control) |
|------------------------|-------|-------|-------|-------------------------------|
| Full                   | 1.000 | 0.859 | 0.619 | 247, 1                        |
| Vitals only            | 1.000 | 0.755 | 0.555 | 38, 0                         |
| Vitals plus engineered | 1.000 | 0.845 | 0.718 | 51, 0                         |
| Vitals plus embeddings | 1.000 | 0.816 | 0.574 | 245, 0                        |

this pipeline was designed for. That evaluation is pending access approval and will be reported separately with the same patient level, confidence interval aware protocol.

## 5 Conclusion

We built and validated a complete multimodal early warning pipeline, and stressed it against the only open access real pre arrest dataset. The machinery is sound and reproducible; the synthetic benchmark is passed; the real data test fails at 6 hours for every model variant we tried, which correctly scopes the problem. The electrocardiogram alone, in a Holter regime with 20 patients, does not carry the 6 hour signal this model needs, and the only robust hint of signal, at 1 hour, comes from engineered rate features. The next and decisive step is the credentialed ICU evaluation with full modalities.

## 6 Reproducibility

Code is available as a single notebook with supporting scripts. Embedding models: ECG FM (repository commit 9f926f19, official Hugging Face release) and PaPaGei S (repository commit 0c537dad, Zenodo record 13983110). Real data: Sudden Cardiac Death Holter Database version 1.0.0 on PhysioNet (open access). All metrics, figures and configuration are exported with the run; seeds are fixed; every expensive step is cached; no patient appears in more than one split.

Table 6: Real data stress test on the Sudden Cardiac Death Holter Database (20 patients, 1829 windows; patient level bootstrap CIs).

| Model                                        | 1 h AUROC            | 6 h AUROC            |
|----------------------------------------------|----------------------|----------------------|
| Zero shot (synthetic trained model)          | 0.338                | 0.447                |
| FEAN, concatenation (baseline)               | 0.632 [0.516, 0.748] | 0.402 [0.344, 0.456] |
| FEAN, identity adversarial                   | 0.697 [0.599, 0.792] | 0.436 [0.368, 0.501] |
| FEAN, identity adversarial + PCGrad          | 0.663 [0.557, 0.781] | 0.459 [0.382, 0.537] |
| FEAN, gated fusion + masks                   | 0.655 [0.546, 0.778] | 0.386 [0.337, 0.435] |
| FEAN, modality attention + masks             | 0.638 [0.519, 0.754] | 0.473 [0.402, 0.538] |
| FEAN, static modality weights + masks        | 0.660 [0.546, 0.776] | 0.394 [0.347, 0.443] |
| Ablation: engineered features only           | 0.767 [0.705, 0.837] | 0.508 [0.461, 0.555] |
| Ablation: embeddings only                    | 0.707 [0.602, 0.812] | 0.422 [0.377, 0.462] |
| Slow risk layer (3 h blocks, QT/TWA/HRV, LR) | 0.426 [0.193, 0.667] | 0.313 [0.190, 0.400] |

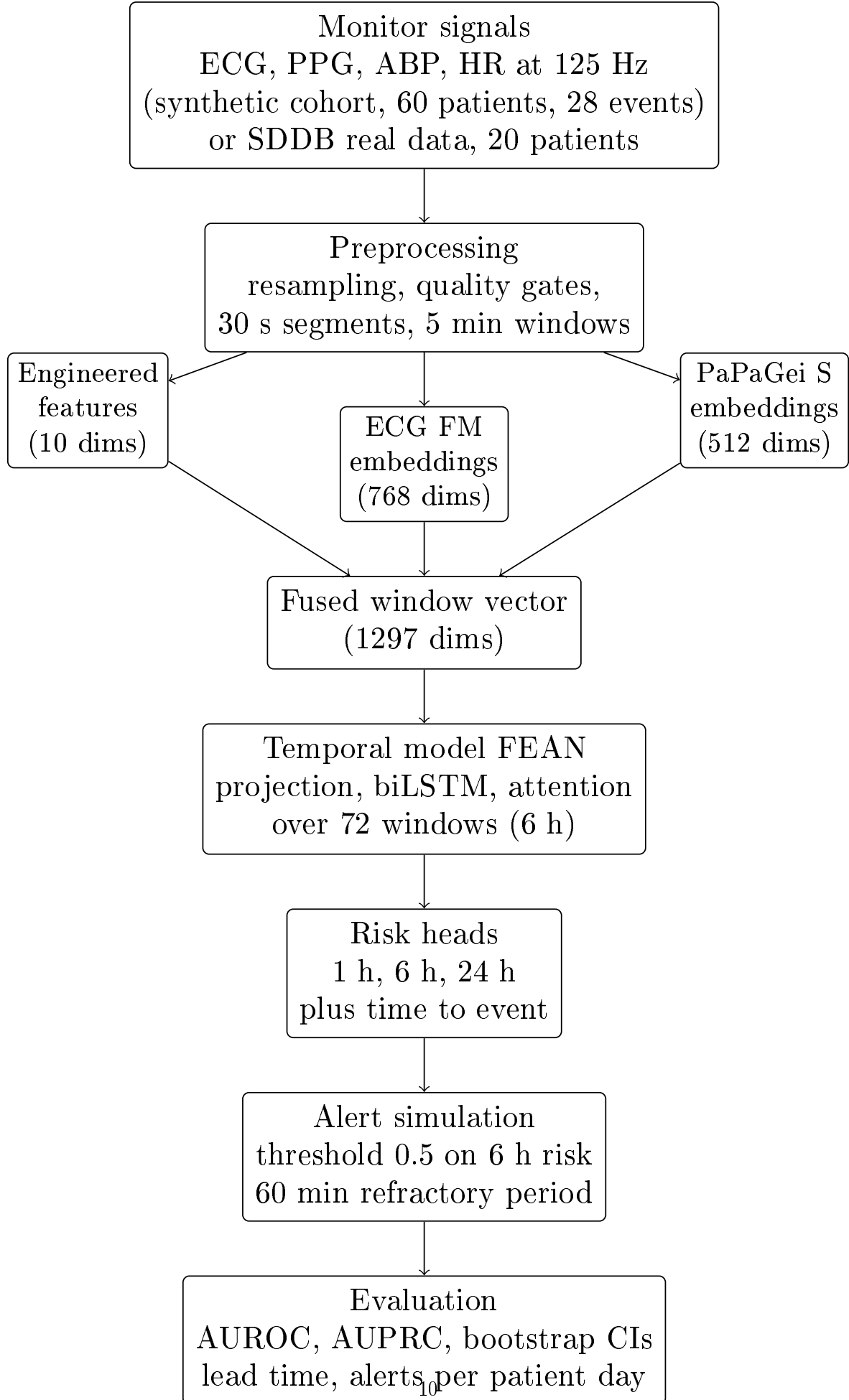


Figure 1: Pipeline overview: signals, preprocessing, three feature blocks, the temporal model FEAN, alert simulation and evaluation.

Synthetic patient 2: pre-arrest drift over the last 12 h

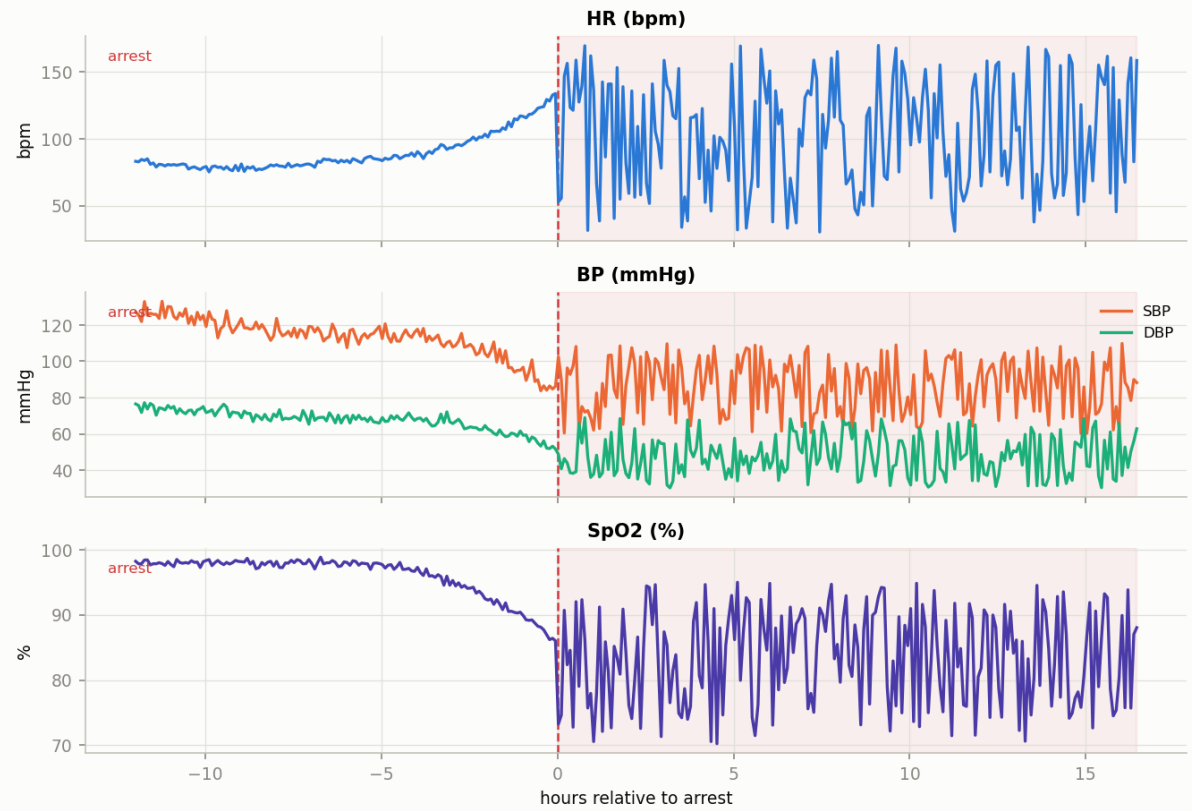


Figure 2: Pre arrest drift of one synthetic case patient over the last 12 hours before arrest (heart rate, blood pressure, saturation). The post arrest region is shaded and excluded downstream.

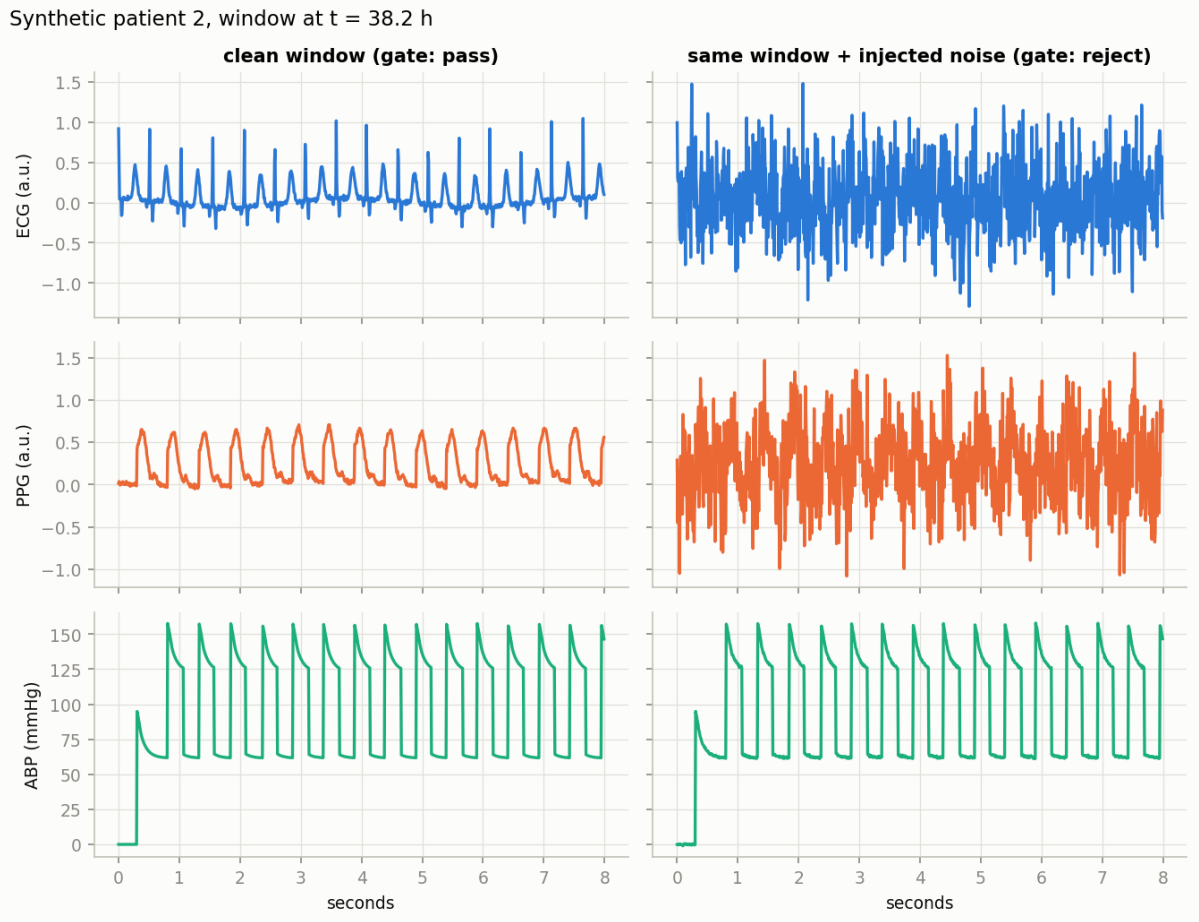


Figure 3: One 30 second window: clean electrocardiogram, photoplethysmogram and arterial pressure (left) versus the same window with an injected noise artifact (right), which the quality gate rejects.

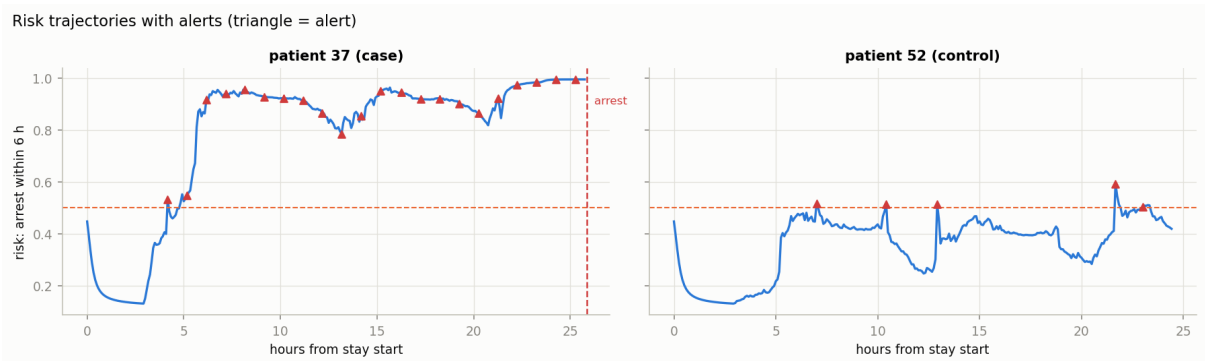


Figure 4: Risk trajectories of the trained pipeline on the two held out patients. Triangles mark alerts, the dashed line is the threshold. The case patient (left) first crosses the threshold about 21 hours before arrest; the control (right) produces a small number of false alarms.

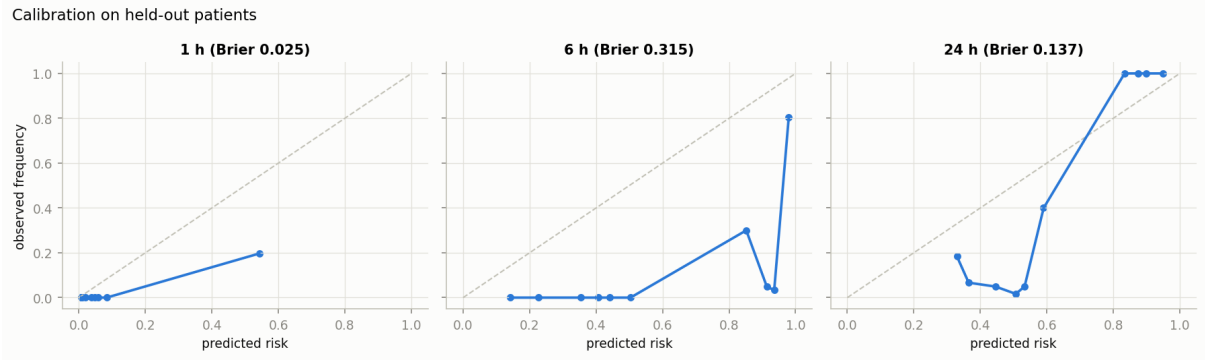


Figure 5: Calibration curves per horizon on the held out patients. Predicted risk versus observed frequency; the diagonal is perfect calibration.

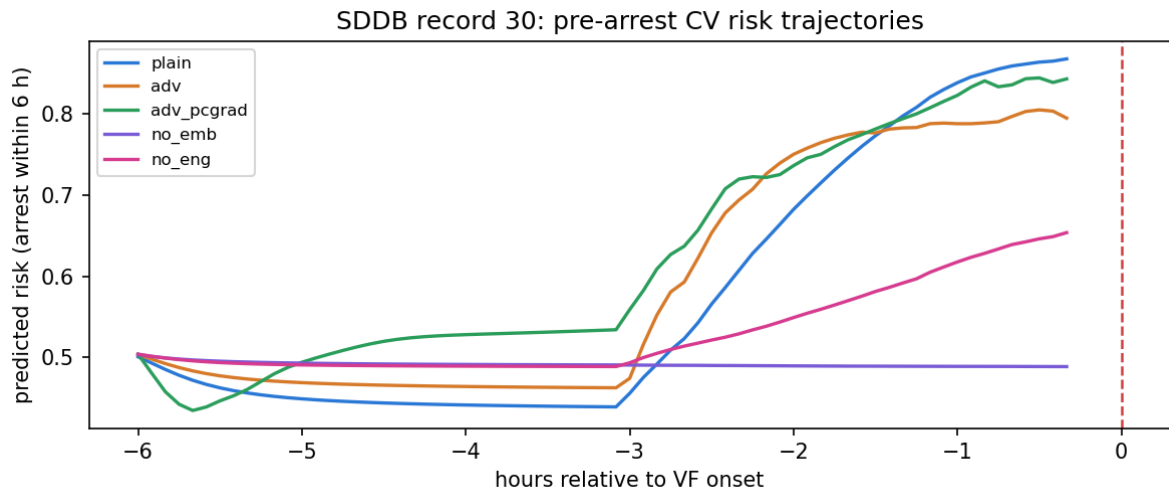


Figure 6: Cross validated risk trajectories for one real patient, hours relative to ventricular fibrillation onset. The predicted risk does not rise before the real event in any variant.

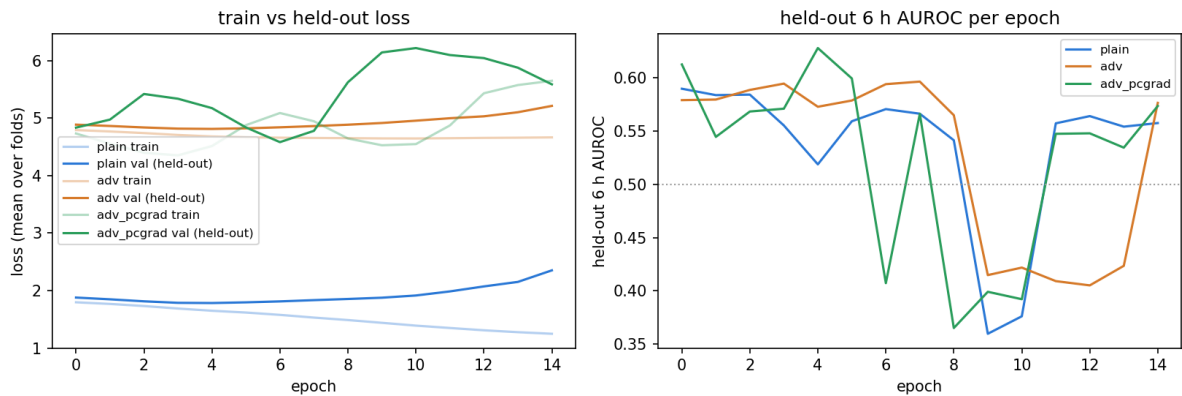


Figure 7: Train versus held out loss per epoch (mean over folds) for the three identity adversarial variants, and held out 6 hour AUROC per epoch.

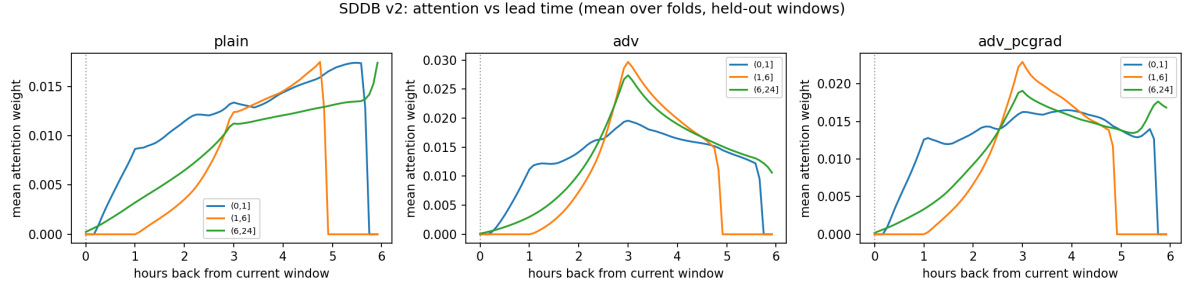


Figure 8: Attention salience by lead time bin (mean over folds): the model spreads attention over history and puts near zero weight on the current window.

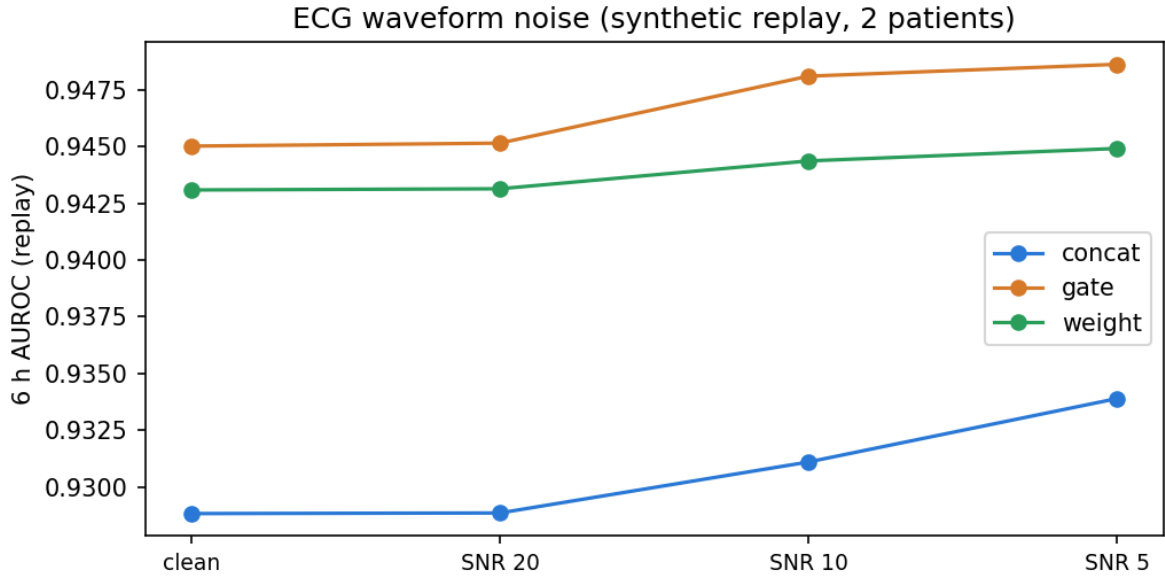


Figure 9: 6 hour AUROC versus ECG waveform noise level (synthetic replay). No degradation down to SNR 5 dB.

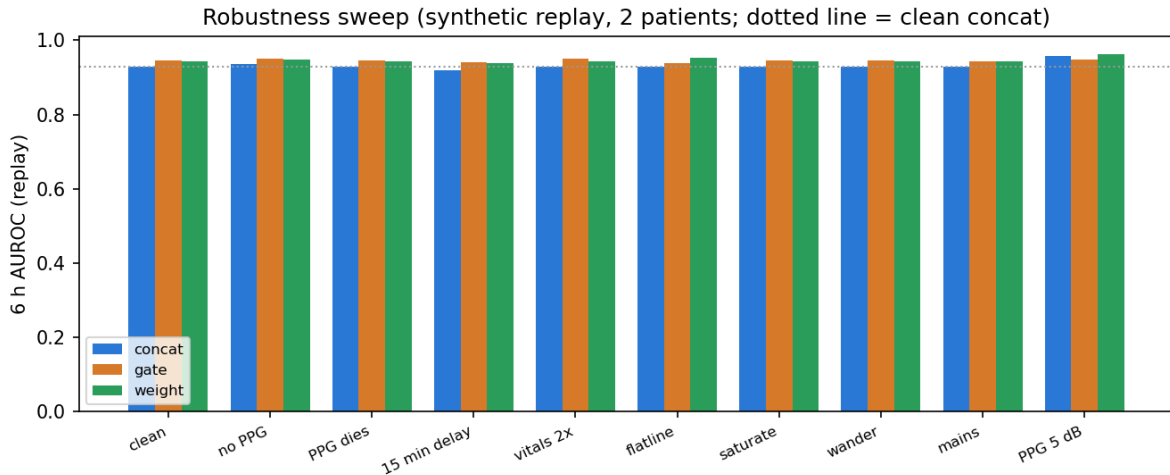


Figure 10: Robustness sweep across all perturbation families (synthetic replay; the dotted line is the clean concatenation score).